Bess Hagan

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Summary

Analytics-focused Computer Science graduate with experience analyzing permitting data, automating reporting, and administering permitting software for a city planning department. Skilled at validating messy data and communicating findings to non-technical audiences. Background in SQL, Python, and machine learning research.

Experience

Planning Specialist *Jul 2026 – Present*
City of Georgetown Planning Department

- Analyzed 129 landscaping review requirements (98 permits) from MGO permitting software in an interactive Excel workbook to measure timeliness: 35% could not be assessed, and 58% of the measurable remainder were late. Cross-validation against a second dataset showed the custom report missed 52% of the projects handled that month; presented findings to the department director and adapted the workbook for Certificate of Appropriateness reviews.
- Produced the department's performance measures report from documented SOPs, consolidating MGO exports with manually maintained data and refreshing Power BI dashboards for internal and public views; determined the source data could not be validated and recommended retiring the report pending a rebuild.
- Automated the biweekly Submittal Report with a macro-enabled Excel template, cutting ~30 minutes of manual work per cycle; validated through an independent run by a colleague and documented in an updated SOP.
- Administer MGO: configured a new Water Utilities review department across two workflows, linked new Unified Development Code checklists in the customer portal, and led the automated case expiration rollout, including cross-department communication and a log tracking extension requests and resubmittal deadlines.
- Authored SOPs for previously undocumented processes, including annual committee roster updates; serve as Historic and Architectural Review Commission (HARC) and Demolition Subcommittee liaison and as the department's first point of contact for applicants and the public.

Procedural Content Generation Researcher *May 2025 – Jul 2025*
Southwestern University (SURF 2025), Georgetown, TX

- Coauthored a peer-reviewed paper on diffusion-based game level generation, presented at the 21st AAAI Conference on AI and Interactive Digital Entertainment (AIIDE-25). https://arxiv.org/abs/2507.00184
- Trained conditional and unconditional UNet diffusion models in PyTorch, implementing cross-entropy loss and early stopping within an existing codebase (https://github.com/schrum2/MarioDiffusion); wrote Python scripts to evaluate and visualize A* solvability metrics across models and batch scripts to automate data preparation, training, and reporting.

Data Researcher *May 2024 – Aug 2024*
University of Illinois Urbana-Champaign (INVITE Institute), Champaign, IL

- Analyzed structured behavioral log data with SQL and Python for ~100 users in a Minecraft-based learning environment to identify markers of student persistence and struggle.
- Built a SQL-to-Python pipeline to extract, clean, and transform multi-table relational event logs into analysis-ready binary datasets, defining 10+ behavioral event types (e.g., puzzle completion state, run and reset counts, Levenshtein edit distance between attempts).
- Performed data validation and consistency checks across datasets and reported cross-system discrepancies to developers.

Projects

E-Commerce Customer Behavior Analysis *Feb 2026 – Mar 2026*

Independent Project

- Analyzed ~14,000 e-commerce orders (public Kaggle dataset) in Excel (pivot tables, charts) to answer five business questions; found three categories drove 81% of revenue from 38% of units, and returning customers placed ~86% of orders at a nearly identical average order value, pointing to retention over discounts, device, or delivery speed as the growth lever.
- Reproduced the full analysis in Python (Pandas, Matplotlib) to validate Excel results and compare workflows; published a report with visualizations for non-technical audiences (https://besshagan.github.io/excel_vs_python).

KNN Classification Project *Nov 2024 – Dec 2024*

Academic Project, Southwestern University

- Built a KNN classifier in Python (Pandas, scikit-learn) on a 61,000-record dataset, handling missing values across 21 features and validating data quality against a synthetic test case before training.
- Performed exploratory data analysis and built a reusable evaluation framework (confusion matrix, ROC and precision-recall curves), achieving 99.5% test accuracy.

Education

B.A. Computer Science | Minor in Data Analytics *Dec 2025*

Southwestern University, Georgetown, TX

- GPA: 4.0, Summa Cum Laude
- Chapter President, Upsilon Pi Epsilon International Honorary Computer Science Society (2024–2025)
- **Selected Coursework:** Database Management, Data Analytics, Statistics, Machine Learning

Skills

Languages: Python, SQL (PostgreSQL), R

Data Analysis and Reporting: Excel (pivot tables, slicers, macros), Power BI, exploratory data analysis, data cleaning and validation, ETL pipelines, relational data modeling, statistical analysis, data visualization

Tools & Systems: Pandas, NumPy, Matplotlib, scikit-learn, Jupyter, Git/GitHub, MGO permitting software

Machine Learning: PyTorch, regression, decision trees, KNN, k-means, diffusion models, model evaluation